Intelligence Amplification:
A Human-Centered Approach to AI-Mediated Education and Professions

JUNAID QADIR
Professor of Computer Engineering,
CSE Department, CENG,
Qatar University



17th World Congress of Bioethics, Doha, Qatar.

Learning Objectives & Conflict of Interest Statement

I do not have an affiliation (financial or otherwise) with any for-profit or not-for-profit organizations.
I have obtained research funding from Facebook Research, QRDI/QNRF/ HEC

At the end of this session, participants will be able to:

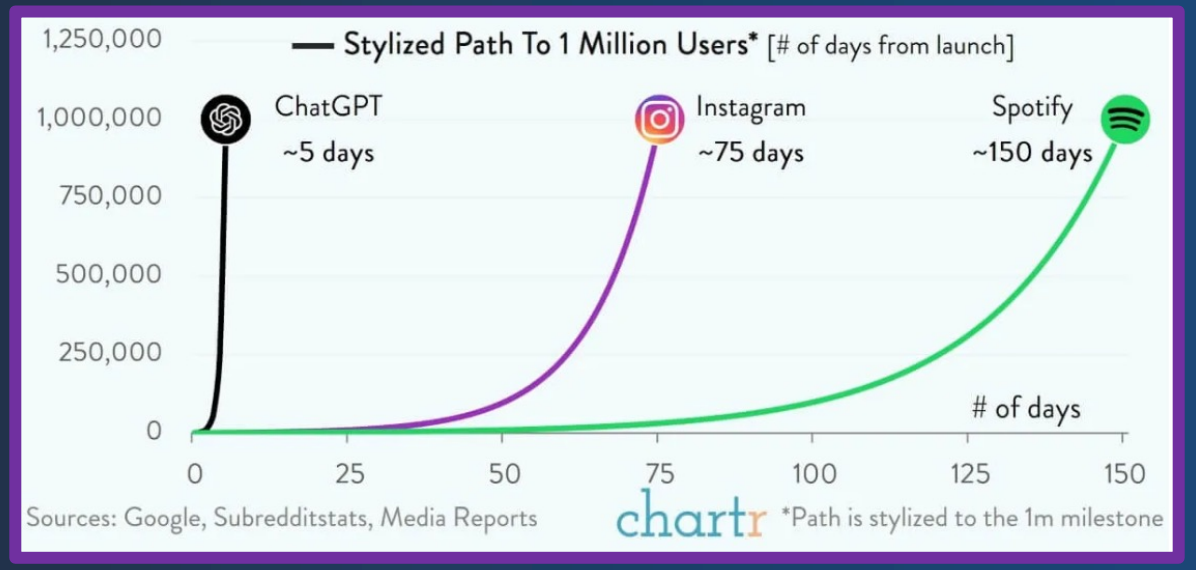
1. Describe the affordances and contradictions of incorporating AI into educational and professional environments (promise & pitfalls).
2. Identify basic strategies and tools to effectively integrate AI technologies into their educational or professional practices.

Outline

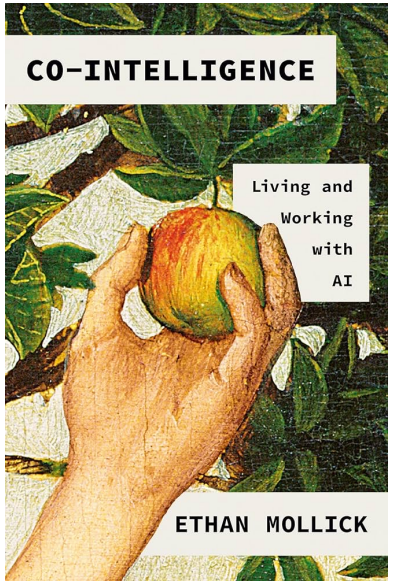
1. Using AI in Education: Introduction
2. Affordances and Contradictions
3. Education for the Exponential AI Era

1

Using AI in Education



The promise and disruption of GenAI



MOTHERBOARD
TECH BY VICE

ChatGPT Is Passing the Tests Required for Medical Licenses and Business Degrees

ChatGPT passes exams from law and business schools

By Samantha Murphy Kelly, CNN Business
Updated 1:35 PM EST, Thu January 26, 2023

Engineering Education in the Era of ChatGPT: Promise and Pitfalls of Generative AI for Education

Junaid Qadir
Department of Computer Science and Engineering, College of Engineering,
Qatar University, Doha, Qatar
jqadir@qu.edu.qa

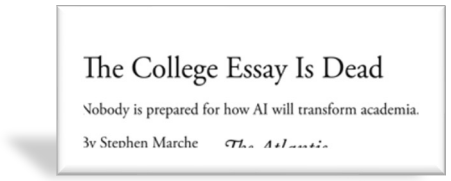
IEEE EDUCON 2023
IEEE Global Engineering Education Conference

Sam Altman ✓
@sama

ChatGPT is incredibly limited, but good enough at some things to create a misleading impression of greatness.

it's a mistake to be relying on it for anything important right now. it's a preview of progress; we have lots of work to do on robustness and truthfulness.

3:11 AM · Dec 11, 2022


The College Essay Is Dead

Nobody is prepared for how AI will transform academia.

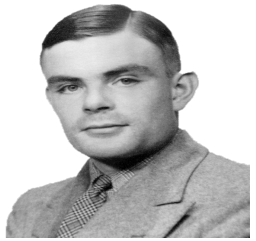
3y Stephen Marche

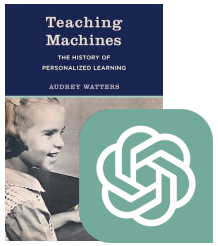
The "Turing Trap": Focusing on Machines

MACHINE-CENTRIC

THE TURING TRAP

CAN MACHINES THINK?





AUTOMATION/ REPLACEMENT FOCUS

THE TEACHING MACHINE TRAP

CAN MACHINES TEACH?

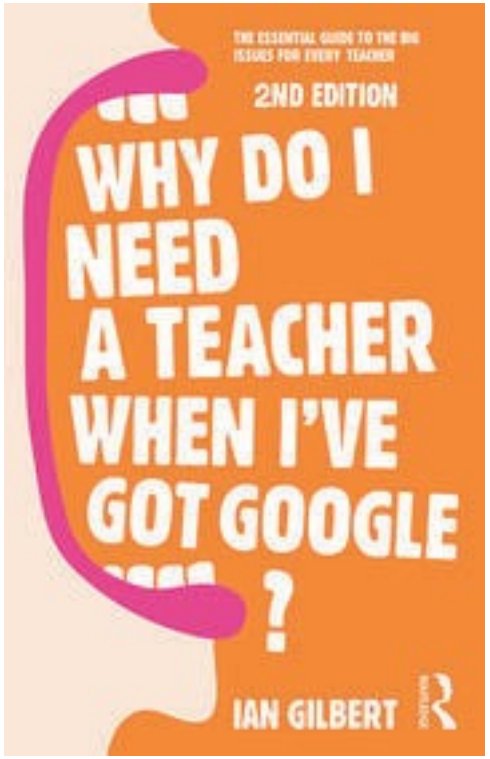
Note the focus on machine intelligence rather than on enhancing or augmenting human intelligence

Now imagine a future, where machines quickly advanced to a level where we could not even equal them or fully understand what they were doing in literally every domain?

Would this new world leave us with a profound motivation gap and the risk we become permanently de-skilled and de-educated?

Do we really need Fully Autonomous Machine Intelligence?

Can learning be automatized and outsourced?



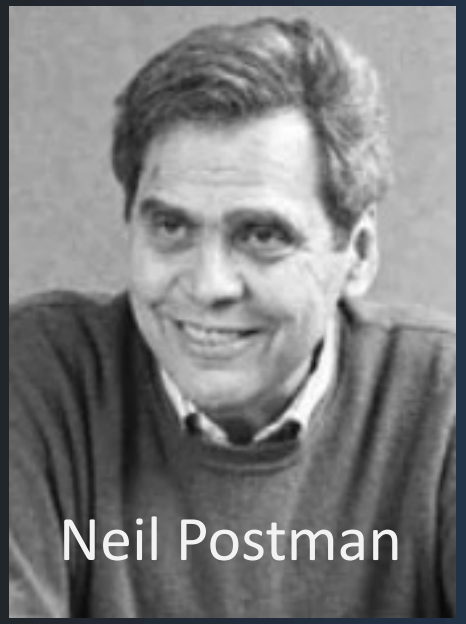
You need to train your mind by doing this and go through the process to self-discover or make your concept your own.

Information Technology and Educational Amnesia

LANGDON WINNER
Rensselaer Polytechnic Institute, Troy, New York, USA

No matter how wonderful the technology (TV, VCR, Internet, Google, ChatGPT), technology alone is not sufficient.Technology can amplify intent and capabilities not substitute for them.

2 Affordances and Contradictions



"Every technology is both a burden and a blessing; not either-or, but this-and-that."

First Idea: Technology giveth and technology taketh away.

Second Idea: Advantages and disadvantages of new technologies are never distributed evenly

Third Idea: Every technology has a philosophy … in how it codifies the world, in which of our senses it amplifies, in which of our emotional and intellectual tendencies it disregards.

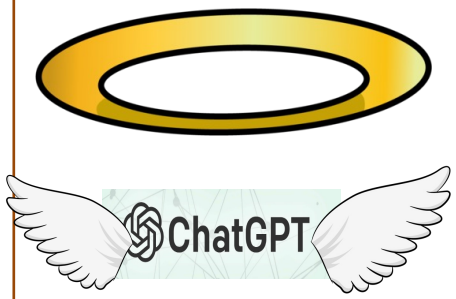
Fourth Idea: Technological change is not additive; it is ecological.

Fifth Idea: Our enthusiasm for technology can turn into a form of idolatry and our belief in its beneficence can be a false absolute.

How we may go wrong with GenAI

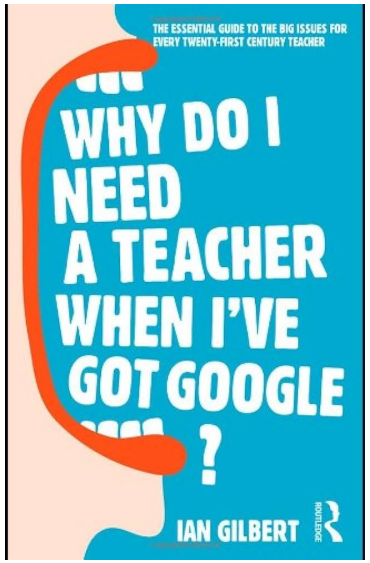
HALO EFFECT

The Gullibility to Believe AI is infallible



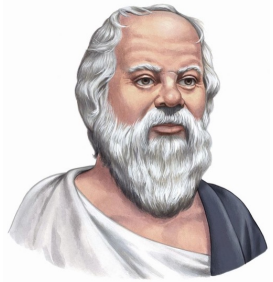
DISINCENTIVES

Why spend time and effort learning when you can just Google/ or use ChatGPT



SOPHISTRY & PRETENTION

Appearances become more important than truth. In Education, the law of harvest applies.





TECHNOLOGICAL OUTSOURCING



AL-GHAZALI AND THE ROBBERS

DEHUMANIZATION

Making the thoughts/ outputs of every student formulaic and cliched



Affordances and Contradictions of AI

June/ July Contradiction

Culture mismatch between workplace and universities

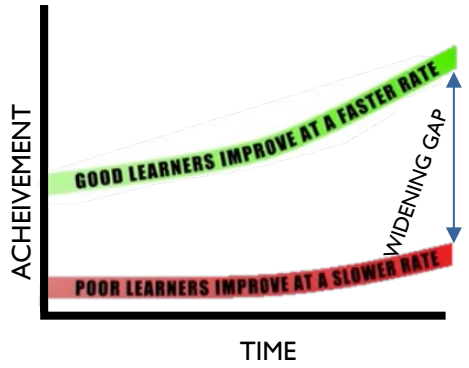


The use of GenAI may be prohibited for students in universities in June.
In July, after graduation, the students are expected to know how to effectively use GenAI in their work.

The rich gets richer contradiction

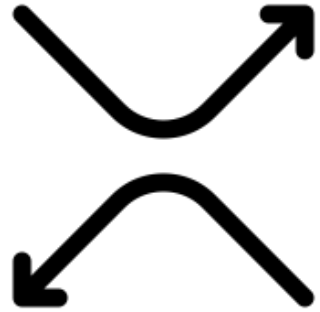
(Matthew Effect)

Those who already know, stand to benefit most from GenAI, as they can prompt, critique and judge the output better



With and Without Contradiction

The people who will be best positioned to benefit from GenAI are those who are already better off without GenAI



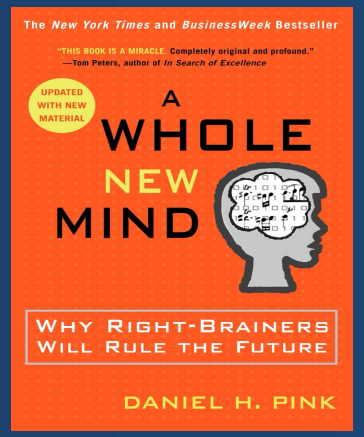
Warschauer, M., Tseng, W., Yim, S., Webster, T., Jacob, S., Du, Q., & Tate, T. (2023). The affordances and contradictions of AI-generated text for writers of English as a second or foreign language. *Journal of Second Language Writing*, *62*, 101071.

3

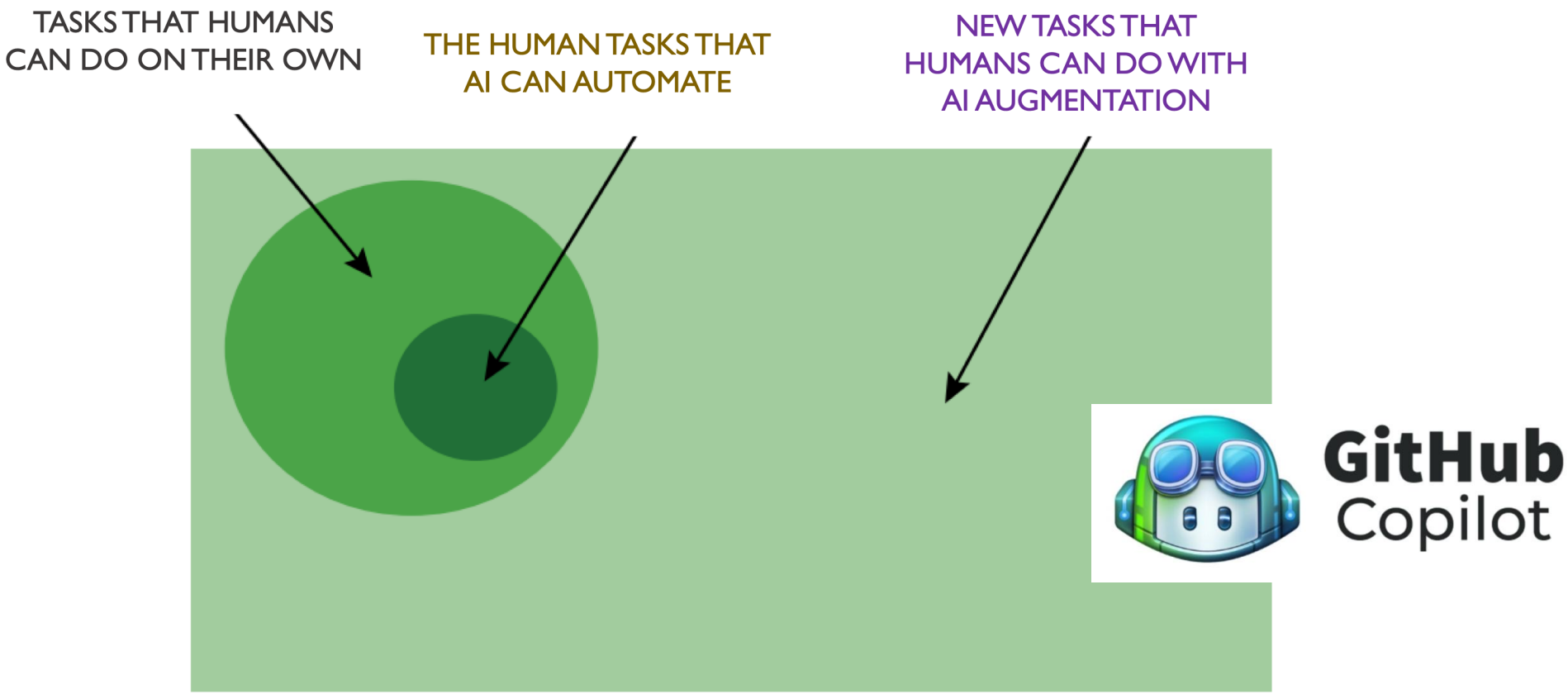
Education for the Exponential AI Era

Developing AI-Proof Learners





Human Centered AI for Intelligence Augmentation



AI for Education, especially, should be:

(1) Human-Centered; and (2) based on Intelligence Augmentation

Holistic Human-Centered View of AI for Education

(focused on what the AI can support humans in)



THE CHARACTER IMPERATIVE



THE UNIQUE HUMAN POTENTIAL REALIZATION IMPERATIVE



INFORMED JUDGEMENT AND CITIZENSHIP



ENGENDERING SHARED PROSPERITY



THE ECONOMIC IMPERATIVE

First Law 

Your work should be no worse than that produced by the AI alone.
(Embrace change and leverage AI)

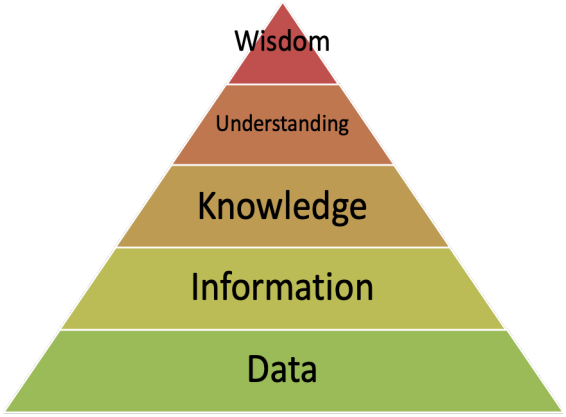
Second Law 

(Corollary of Law 1)
If you don't want to be made irrelevant, maintain your competitive advantage
(Become **AI Proof** and maintain some competitive advantage)

Third Law 

After becoming users of AI, don't let your inherent unaided capabilities drop
(Don't outsource your thinking muscle allowing it to atrophy]

What would educating for wisdom look like?



Wisdom emerges from an understanding of the historical, social, and cultural context. It cannot be completely programmed into logic, mathematics, models, or code

CRITICAL THINKING

—Knowing the context

—Knowing the pitfalls of rules/model/methods

—Ability to engage in informed judgement

—Knowing the limits of your knowledge

الجهل البسيط

الجهل المركّب

DUNNING-KRUGER

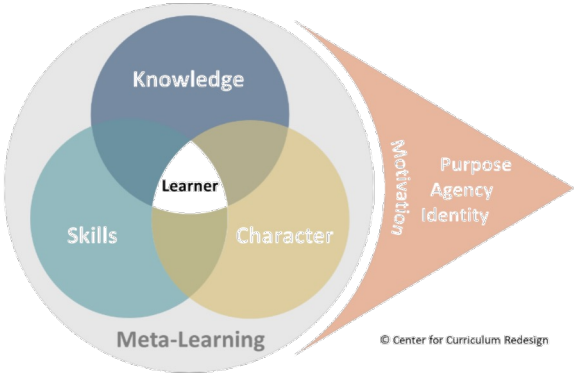
HOLISTIC THINKING

—Multidimensional/ Multidisciplinary Thinking

—Apart from Cognitive, Focus on Character, Skills, & Metacognition

—Intelligence requires both reckoning & judgement

—Dialogic Thinking

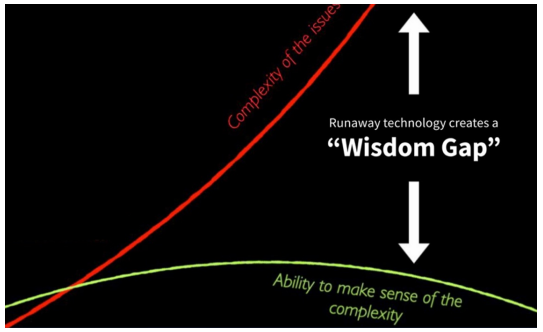


PURPOSEFUL USE OF AI

— Knowing which ends/ objectives/ purpose to pursue and prioritize

— Importance of Human Skills and Character

—Pro-Human & Pro-Social



Conclusion

Our goal in education and professions should focus on:

- *Empowering with AI: Utilize AI as a tool to enhance human capabilities, promoting innovation, creativity, and shared prosperity.*
- *Mitigating AI Risks: Be proactive in identifying and mitigating potential harms, such as AI colonialism and dehumanization.*

The focus should be on the synergy between human abilities and technology, highlighting what humans can achieve with and without AI, rather than on the capabilities of algorithms alone.

THANK YOU!

jqadir@qu.edu.qa